

Hotel Analytics and Revenue Management

A sample knowledge document for a local Retrieval-Augmented Generation (RAG) system

This document introduces common hotel performance measures and explains how they can be used to understand occupancy, room pricing, revenue, reservations, and cancellations. It is designed as a text-based PDF that can be loaded, chunked, embedded, stored in a vector database, and queried by a RAG application.

1. Occupancy Rate

Occupancy rate measures the percentage of available rooms that are occupied during a specific period. It is calculated by dividing the number of rooms sold by the number of rooms available and multiplying the result by 100. For example, if a hotel has 100 available rooms and sells 80 of them, the occupancy rate is 80 percent.

Occupancy is useful for understanding demand, but a high occupancy rate does not automatically mean that the hotel is maximizing revenue. A hotel can sell many rooms at prices that are too low. Managers therefore normally examine occupancy together with pricing and revenue measures.

2. Average Daily Rate (ADR)

Average Daily Rate, usually called ADR, represents the average room revenue earned for each room sold. ADR is calculated as room revenue divided by the number of rooms sold. If a hotel earns EUR 12,000 in room revenue from 80 sold rooms, its ADR is EUR 150.

ADR helps a hotel understand its average selling price. An increase in ADR can improve revenue when demand remains strong. However, setting prices too high may reduce bookings and occupancy, so pricing decisions should consider demand, seasonality, competitors, events, and customer segments.

3. Revenue per Available Room (RevPAR)

Revenue per Available Room, or RevPAR, combines occupancy and room pricing into one performance measure. One common formula is $\text{RevPAR} = \text{ADR} \times \text{occupancy rate}$. It can also be calculated as total room revenue divided by the total number of available rooms.

For example, an ADR of EUR 150 with an occupancy rate of 80 percent produces RevPAR of EUR 120. RevPAR is useful because it reflects both the hotel's ability to fill rooms and its ability to achieve an effective room rate.

4. Reservations and Booking Patterns

Reservation data can reveal when guests book, which channels they use, how long they stay, and how far in advance they make reservations. Important fields may include booking date, arrival date, departure date, room type, booking channel, guest segment, number of guests, and booking status.

Booking lead time is the number of days between the reservation date and the arrival date. Understanding lead time can help a hotel adjust prices and promotions. A hotel may also compare

direct bookings with bookings from online travel agencies and other distribution channels.

5. Cancellations

A cancellation occurs when a confirmed reservation is withdrawn before the guest stays at the hotel. Cancellation rate can be calculated as the number of cancelled reservations divided by the total number of reservations, multiplied by 100.

High cancellation rates can make demand forecasting more difficult and can leave rooms unsold. Hotels can analyse cancellations by booking channel, customer segment, room type, lead time, and season. Cancellation policies, deposits, flexible rates, and non-refundable rates can influence cancellation behaviour.

6. Revenue Analysis

Hotel revenue analysis examines how much money the property earns and where that revenue comes from. Room revenue is usually a major component, but hotels may also earn revenue from food and beverage, events, spa services, parking, and other services.

Revenue should be analysed over time to identify trends. Monthly or weekly comparisons can reveal seasonality, growth, or declining performance. Analysts may compare actual revenue with previous periods, budgets, or targets. Revenue alone should not be interpreted without considering occupancy, ADR, costs, and available capacity.

7. Seasonality and Demand

Hotel demand often changes by season, day of the week, holidays, conferences, concerts, trade fairs, and other local events. A property may experience strong weekend leisure demand but weaker weekday demand, or the opposite for a business-oriented hotel.

Historical reservation data can help identify recurring demand patterns. Revenue managers use these patterns to make pricing and inventory decisions. During periods of high expected demand, a hotel may increase room rates or apply stricter booking conditions. During low-demand periods, promotions may be used to stimulate bookings.

8. Hotel Performance Dashboard

A hotel performance dashboard presents important measures in a form that decision makers can understand quickly. Typical key performance indicators include total revenue, occupancy rate, ADR, RevPAR, reservations, cancellations, and average length of stay.

Charts can show revenue trends over time, booking channel performance, room-type performance, and cancellation patterns. Filters can allow the user to focus on particular dates, properties, customer segments, or room types. A dashboard should be designed around the needs of its intended user rather than displaying every available metric.

9. Using Analytics for Decisions

Hotel analytics supports decisions by converting operational data into useful information. For example, a manager may notice that occupancy is high but RevPAR is weak. This could indicate that rooms are being sold at rates that are too low. Another analysis may show that a particular booking channel generates many reservations but also has a high cancellation rate.

Good analysis should connect metrics to business questions. Instead of only reporting what happened, analysts should investigate why it happened and what action may improve future performance.

10. Example Questions for a RAG System

A RAG chatbot using this document should be able to answer questions such as: What is ADR? How is RevPAR calculated? What does occupancy rate measure? Why are cancellations important? What is booking lead time? Which KPIs could appear on a hotel performance dashboard? How can seasonality influence hotel pricing?

Because the RAG system retrieves relevant passages before generating an answer, the language model can ground its response in this document rather than relying only on its general training knowledge.

Key Formula Summary

Metric	Formula
Occupancy Rate	$(\text{Rooms Sold} / \text{Rooms Available}) \times 100$
ADR	$\text{Room Revenue} / \text{Rooms Sold}$
RevPAR	$\text{ADR} \times \text{Occupancy Rate}$, or $\text{Room Revenue} / \text{Available Rooms}$
Cancellation Rate	$(\text{Cancelled Reservations} / \text{Total Reservations}) \times 100$